

A new bound for the Furstenberg–Sárközy theorem using the van der Corput property

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Abstract

We show that there is an absolute constant $c > 0$ such that if $A \subseteq \mathbb{N} \cap [1, N]$ has no nonzero square difference, then

$$|A| \leq N \exp(-c\sqrt{\log N \log \log N}),$$

improving upon a recent result of Green and Sawhney. The proof exploits a quantitative version of the van der Corput property with signed coefficients and builds on previous constructions of Slijepčević, Slijepčević–Ninčević, and Fan–Lott. The proof of the upper bound is elementary and self-contained. We also prove a matching lower bound for the constant coefficient of any van der Corput witness for squares, showing that our quantitative van der Corput bound is sharp up to the constant c .

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1 Introduction

For a set $F \subseteq \mathbb{Z}$ and $N > 1$, write

$$D(F, N) := \max\{|A| : A \subseteq [N] \text{ and } (A - A) \cap F \subseteq \{0\}\},$$

where $[N] := \mathbb{N} \cap [1, N]$. Here, we refer to F as the set of *forbidden differences*.

The Furstenberg–Sárközy theorem asserts that $D(\{n^2 : n \in \mathbb{N}\}, N) = o(N)$, answering a question of Lovász. This was proved independently by Furstenberg [Fur77] using ergodic theory and by Sárközy [Sár78a, Sár78b] using the Hardy–Littlewood circle method and the density increment method of Roth; Sárközy’s proof gives the quantitative bound

$$D(\{n^2 : n \in \mathbb{N}\}, N) \ll N(\log \log N)^{2/3}(\log N)^{-1/3}.$$

Pintz–Steiger–Szemerédi [PSS88] introduced a more elaborate double iteration strategy to obtain a bound of the form

$$D(\{n^2 : n \in \mathbb{N}\}, N) \ll N(\log N)^{-c_1 \log \log \log \log N}$$

with $c_1 = 1/12$, and Bloom–Maynard [BM22] later strengthened this to

$$D(\{n^2 : n \in \mathbb{N}\}, N) \ll N(\log N)^{-c_2 \log \log \log N}$$

where $c_2 > 0$ is an absolute constant. Green and Sawhney [GS24, GS25] developed a new “arithmetic level- d ” inequality (building on hypercontractivity estimates of Keller–Lifshitz–Marcus [KLM26] and Keevash–Lifshitz–Long–Minzer [KLLM24]) and pushed the density increment strategy to obtain the following quasipolynomial bound

$$D(\{n^2 : n \in \mathbb{N}\}, N) \ll N \exp(-c_3(\log N)^{1/2}), \tag{1}$$

for some constant $c_3 > 0$. They also construct an example which they interpret as ruling out bounds stronger than $N \exp(-c(\log N)^{3/4})$ by the density increment method, “at least in anything like the form in the present paper (and in previous works)” [GS25, Section 9].

Another approach for bounding $D(F, N)$ is the quantitative van der Corput method. For a positive integer N and a small parameter $\delta > 0$, one seeks a real trigonometric polynomial

$$P(\theta) = a_0 + \sum_{n=1}^N a_n \cos(2\pi n\theta)$$

with the following properties:

- (i) $P(\theta) \geq 0$ for every $\theta \in \mathbb{T}$;
- (ii) $P(0) = 1$;
- (iii) $a_n = 0$ whenever $1 \leq n \leq N$ and $n \notin F \cup (-F)$;
- (iv) $0 \leq a_0 \leq \delta$.

These properties imply $D(F, N) \ll N\delta$. This implication goes back to Kamae–Mendès France [KMF78] and Ruzsa [Ruz84a]; we give the proof for squares in Appendix A. Montgomery [Mon94] gives a useful account of the method. A recent breakthrough of Green [Gre24] gives a demonstration of the strength of this method: by proving a power-saving form of the van der Corput property for the shifted primes, he obtained the first power-saving bound for Sárközy’s theorem for differences of the form $p - 1$ with p prime.

For squares in $[N]$, Slijepčević [Sli10] constructed such polynomials with constant coefficient $O((\log N)^{-1/3})$. Slijepčević's construction for squares averages over a carefully chosen finite collection of dilations so that their real major-arc contributions combine to give a uniform lower bound. Ninčević and Slijepčević [NS14] subsequently used averaging over dilations for odd integer polynomials of degree $k \geq 3$, obtaining constant coefficient $O((\log N)^{-1/k})$.

More recently, Fan and Lott [FL26] used the quantitative van der Corput method to prove that if $A \subseteq [N]$ and $A - A$ does not contain any positive integer of the form $x^2 + y^2$ with $x, y \in \mathbb{Z}$, then $|A| \ll_\epsilon N^{7/8+\epsilon}$ for every fixed $\epsilon > 0$. The constructed cosine polynomials also have nonnegative coefficients. The key idea there is to average over dilations in the style of Ninčević–Slijepčević and then carefully select smooth weights so that the complete sum and oscillatory integral lie in the same sector in the complex plane defined by $|\arg z| \leq \pi/4$, resulting in the positivity of the real part of their product.

Building on these ideas, we prove the following quantitative van der Corput property for squares using cosine polynomials with signed coefficients.

Theorem 1.1. *For every sufficiently large integer N there is a real trigonometric polynomial*

$$P(\theta) = a_0 + \sum_{1 \leq n \leq \sqrt{N}} a_n \cos(2\pi n^2 \theta)$$

such that $P \geq 0$, $P(0) = 1$, and

$$0 \leq a_0 \leq \exp(-c\sqrt{\log N \log \log N}),$$

where $c > 0$ is some constant. The coefficients a_n are allowed to have either sign.

Corollary 1.2. *There is some absolute constant $c > 0$ such that every set $A \subseteq \{1, \dots, N\}$ with no nonzero square differences satisfies*

$$|A| \leq N \exp(-c\sqrt{\log N \log \log N})$$

whenever N is sufficiently large.

The use of signed coefficients in Theorem 1.1 is essential. For $F = \{n^2 : n \in \mathbb{N}\}$, a result of Ruzsa, as reported by Slijepčević [Sli10, Section 2], shows that any polynomial satisfying (i)–(iii) with nonnegative coefficients a_n must have $a_0 \gg 1/\log N$. Thus the bound in Theorem 1.1 requires some negative coefficients. On the other hand, the following matching lower bound shows that Theorem 1.1 is essentially best possible.

Theorem 1.3. *There is an absolute constant $C > 0$ such that, for every sufficiently large integer N , every real trigonometric polynomial*

$$P(\theta) = a_0 + \sum_{1 \leq n \leq \sqrt{N}} a_n \cos(2\pi n^2 \theta)$$

satisfying $P \geq 0$ and $P(0) = 1$ has

$$a_0 \geq \exp(-C\sqrt{\log N \log \log N}).$$

The coefficients a_n may have either sign.

Consequently, Theorem 1.1 is sharp up to the constant in the exponent among the cosine polynomials satisfying (i)–(iii).

Using the quantitative van der Corput method with a different construction, it is also possible to prove that if a subset $A \subseteq [N]$ lacks cube differences, then $|A| \ll N^{1-c}$ for some constant $c > 0$.

This will be the main result of a forthcoming paper. The constant $c > 0$ can be made explicit and reasonable.

We also mention two further applications of the quantitative van der Corput method due separately to Kalogirou, Lott, Magyar, and Singha Roy [KLMSR26] on sums of an even number of equal odd powers of positive integers, and to Younis [You26] on bracket quadratics. Both obtain power-saving bounds.

For lower bounds, Ruzsa [Ruz84b] constructed sets with no nonzero square difference of size $\gg N^{0.733077\dots}$. Lewko [Lew15] and, independently, Beigel and Gasarch [BG08] improved the exponent to $\frac{1}{2}(1 + \log 12 / \log 205) = 0.7334117\dots$. Krachun [Kra26] broke the natural $3/4$ barrier of this approach, obtaining the exponent $0.7527964558\dots$. Several further AI-generated constructions have also been claimed [Jon26a, Nas26, Jon26b]. (See [DHH⁺26] for more background on lower-bound constructions.)

Generalizing the Furstenberg–Sárközy problem on squares, one may also consider sets $A \subseteq [N]$ whose difference sets avoid nonzero integers of the form $h(n)$ with $n \in \mathbb{N}$, where $h \in \mathbb{Z}[x]$ is a fixed intersective polynomial of degree at least two. Due to a simple congruence obstruction, this is the maximal class of polynomials for which a Furstenberg–Sárközy-type conclusion could hold. The state of the art is due to Adajar, Agrawal, Choudhuri, Chuah, Fan, Hegde, Lott, Nandakumar, and Ponagandla [AAC⁺26], who combined the method of Green and Sawhney with novel exponential sum estimates in the style of Rice [Ric19] to show that for every fixed $\mu \in (0, 1/2)$,

$$D(h(\mathbb{N}), N) \ll_{h,\mu} N \exp(-c_h(\log N)^\mu).$$

Doyle and Rice [DR21] obtained similar bounds for a broad class of intersective multivariable polynomials $P \in \mathbb{Z}[x_1, \dots, x_\ell]$ of degree at least two.

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3 AI disclosure

The authors used a large language model (LLM) as a sounding board while developing the proof. In particular, they used the LLM to experiment with van der Corput constructions for squares using signed coefficients, and the LLM had access to the authors’ paper on sums of two squares [FL26]. This led to the definition of λ_v , which was then combined with the ideas in [FL26] to develop the argument in this manuscript. The proof of Theorem 1.3 was discovered during later conversations with the LLM. The LLM was also used for copyediting, LaTeX assistance, and literature search, but the manuscript was prepared and verified by the authors, who take responsibility for the contents of the proof.

4 Notation and conventions

We collect here the notation and conventions used throughout the paper.

- For a positive integer N , we write

$$[N] = \{1, 2, \dots, N\}.$$

If A is a set of integers, then

$$A - A = \{a - a' : a, a' \in A\}.$$

We write 1_A for the indicator function of A .

- Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For $t \in \mathbb{R}$, we put $e(t) = e^{2\pi it}$.
- All divisors and divisor indices are positive unless stated otherwise. The notation (a, q) denotes the greatest common divisor, and μ is the Möbius function.
- Given a nonzero integer n and a positive real number y , we say that n is *y-smooth* if every prime factor of n is at most y and *y-rough* if every prime factor of n is strictly greater than y .
- We write $\mathcal{S}(\mathbb{R})$ for the Schwartz class of complex-valued functions. So $f \in \mathcal{S}(\mathbb{R})$ means that f is smooth and that for every pair of nonnegative integers $\alpha, \gamma \in \mathbb{Z}_{\geq 0}$,

$$\sup_{x \in \mathbb{R}} |x^\alpha f^{(\gamma)}(x)| < \infty.$$

Equivalently, f and all of its derivatives decay faster than any power.

- For $f \in \mathcal{S}(\mathbb{R})$, we have the Fourier transform:

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e(-x\xi) dx, \quad \xi \in \mathbb{R}.$$

All integrals over $\mathbb{R}$ are with respect to Lebesgue measure.

- We also have the Poisson summation formula: if $f \in \mathcal{S}(\mathbb{R})$, then

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{\ell \in \mathbb{Z}} \widehat{f}(\ell).$$

We will also use the scaled form

$$\sum_{n \in \mathbb{Z}} f\left(\frac{n}{H}\right) = H \sum_{\ell \in \mathbb{Z}} \widehat{f}(H\ell)$$

for all $H > 0$, which follows by applying Poisson summation to the function $x \mapsto f(x/H)$.

- We will also use Poisson summation after splitting into residue classes. If $f \in \mathcal{S}(\mathbb{R})$, $q \in \mathbb{N}$, and $r \in \mathbb{Z}/q\mathbb{Z}$, then

$$\sum_{\substack{n \in \mathbb{Z} \\ n \equiv r \pmod{q}}} f(n) = q^{-1} \sum_{\ell \in \mathbb{Z}} e\left(\frac{r\ell}{q}\right) \widehat{f}\left(\frac{\ell}{q}\right), \quad (2)$$

which follows by applying Poisson summation to the function $x \mapsto f(qx + r)$.

- On $\mathbb{Z}/M\mathbb{Z}$, we use the discrete Fourier transform

$$\widehat{f}(\xi) = \sum_{x \pmod{M}} f(x) e\left(\frac{x\xi}{M}\right), \quad \xi \in \mathbb{Z}/M\mathbb{Z},$$

where $f : \mathbb{Z}/M\mathbb{Z} \rightarrow \mathbb{C}$ is any function. With this convention, Parseval's identity is

$$\frac{1}{M} \sum_{\xi \pmod{M}} |\widehat{f}(\xi)|^2 = \sum_{x \pmod{M}} |f(x)|^2.$$

- We use the Landau–Vinogradov notation in the usual way. Thus $X \ll Y$ and $X = O(Y)$ both mean that $|X| \leq CY$ for some constant $C > 0$. Subscripts indicate the allowed dependence of the implicit constant. All logarithms are natural. The constant $c_0 > 0$ below is chosen sufficiently small and then fixed, and N is sufficiently large in terms of this choice.

- We write $X \asymp Y$ to mean $X \ll Y$ and $Y \ll X$.
- For every $t \in \mathbb{R}$, we denote by $\lfloor t \rfloor$ the greatest integer not exceeding t , and by $\lceil t \rceil$ the least integer no less than t .
- We write $\operatorname{Re} z$ and $\arg z$ for the real part and argument of a complex number z . Whenever a principal branch is mentioned, $\arg z$ is taken in $(-\pi, \pi]$. In particular, $z^{-1/2}$ denotes the inverse square root with respect to the principal branch.

5 Sketch of the proof

For a sufficiently large integer N , put

$$H = \frac{\sqrt{N}}{\log N}, \quad Y = \sqrt{\frac{\log H}{\log \log H}}. \quad (3)$$

The constant $c_0 > 0$ is sufficiently small and fixed throughout.

For each positive integer d , we may decompose it uniquely as $d = d_1 d_2$, where d_1 is Y^2 -smooth and d_2 is Y^2 -rough. For $0 \leq u \leq 1$, define

$$\beta_d(u) = d_1^{-c_0/Y} d_2^{-u/(2 \log Y)}, \quad b_d = c_0 Y \int_0^1 \beta_d(u) u^{c_0 Y - 1} du. \quad (4)$$

Furthermore, define the signed weights

$$\lambda_v = \sum_{\substack{d \leq H^{1/8} \\ v|d}} \mu(d/v) b_d = \sum_{k \leq H^{1/8}/v} \mu(k) b_{vk} \quad (1 \leq v \leq H^{1/8}). \quad (5)$$

By Möbius inversion,

$$b_d = \sum_{\substack{v \leq H^{1/8} \\ d|v}} \lambda_v \quad (1 \leq d \leq H^{1/8}). \quad (6)$$

Put

$$\mathcal{B}_H(\theta) = \sum_{v \leq H^{1/8}} v \lambda_v \sum_{\substack{x \in \mathbb{Z} \\ 0 < v^2 x^2 \leq N}} e^{-v^2 x^2 / H^2} e(\theta v^2 x^2). \quad (7)$$

The polynomial in Theorem 1.1 will be obtained by adding a constant to $\operatorname{Re} \mathcal{B}_H$ and normalizing. The following estimates ensure that the resulting polynomial is nonnegative and has a small constant coefficient. The construction should be compared with that of [FL26]. Both use the dilation by v^2 and Gaussian weights, while the main distinction is the choice of the signed coefficients λ_v .

Proposition 5.1. *For a sufficiently small choice of c_0 , we have*

$$\mathcal{B}_H(0) = \sqrt{\pi} H + O(H^{3/4}). \quad (8)$$

In particular, $\mathcal{B}_H(0) > 0$ and $\mathcal{B}_H(0) \asymp H$ for sufficiently large H . Moreover, there are constants $c, C > 0$ such that

$$\operatorname{Re} \mathcal{B}_H(\theta) \geq -CH^{1-c/Y} \quad (\theta \in \mathbb{T}). \quad (9)$$

Proof of Theorem 1.1 assuming Proposition 5.1. Let $c, C > 0$ be as in Proposition 5.1, and put

$$P(\theta) = \frac{CH^{1-c/Y} + \operatorname{Re} \mathcal{B}_H(\theta)}{CH^{1-c/Y} + \mathcal{B}_H(0)}.$$

By Proposition 5.1, we have $P \geq 0$ and $P(0) = 1$. Pairing x and $-x$ in (7) and grouping the terms with $v|x| = n$, we obtain

$$P(\theta) = a_0 + \sum_{1 \leq n \leq \sqrt{N}} a_n \cos(2\pi n^2 \theta),$$

where

$$a_0 = \frac{CH^{1-c/Y}}{CH^{1-c/Y} + \mathcal{B}_H(0)}$$

and

$$a_n = \frac{2e^{-n^2/H^2}}{CH^{1-c/Y} + \mathcal{B}_H(0)} \sum_{\substack{v|n \\ v \leq H^{1/8}}} v \lambda_v \quad (1 \leq n \leq \sqrt{N}).$$

Since $\mathcal{B}_H(0) \asymp H$, we have

$$0 < a_0 \ll H^{-c/Y}.$$

Finally, $\log H = \frac{1}{2} \log N - \log \log N$, so

$$\frac{\log H}{Y} = \sqrt{\log H \log \log H} \asymp \sqrt{\log N \log \log N}.$$

This proves the theorem after decreasing the constant in the exponent. $\square$

It therefore remains to prove Proposition 5.1. Define the quadratic Gauss sum and its signed average by

$$G_q(a) = \sum_{x \bmod q} e(ax^2/q), \quad C_q(a) = \frac{1}{q} \sum_{v \leq H^{1/8}} \lambda_v G_q(av^2).$$

For $\theta = a/q + \eta$, where $(a, q) = 1$, $q \leq H$ and $|\eta| \leq 2/(qH)$, Section 7 proves

$$\mathcal{B}_H(\theta) = \left(\int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx \right) C_q(a) + O(H^{3/4}). \quad (10)$$

This estimate follows from Poisson summation and is strong enough that a Farey arc decomposition of $\mathbb{T}$ suffices, so no major and minor arc decomposition is needed. A standard calculation gives

$$\int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx = \frac{\sqrt{\pi} H}{\sqrt{1 - 2\pi i \eta H^2}}$$

Thus the oscillatory integral in the main term of the estimate has argument in $(-\pi/4, \pi/4)$. Put

$$D = D(q) = \prod_{p^m \parallel q} p^{\lceil m/2 \rceil},$$

so that $D \mid v$ is equivalent to $q \mid v^2$. Since $b_D > 0$, it suffices to prove the estimate

$$C_q(a) = b_D(1 + O(c_0)) + O(H^{-c/Y})$$

of Proposition 6.1. Indeed, for sufficiently small c_0 , the term $b_D(1 + O(c_0))$ has argument in $(-\pi/4, \pi/4)$, so its product with the oscillatory integral has argument in $(-\pi/2, \pi/2)$, and hence has positive real part. This type of sector strategy already appears in [FL26].

Now we give the idea behind the estimate for $C_q(a)$, starting with the case of a prime power. These calculations should be compared to Section 4 of [FL26]. For $q = p^m$, put $k = \lceil m/2 \rceil$. If $p^k \mid v$, then $p^m \mid v^2$, so $G_{p^m}(av^2)/p^m = 1$. If $p^j \parallel v$ with $j < k$, we have

$$G_{p^m}(av^2)/p^m = G_{p^{m-2j}}(a)/p^{m-2j}$$

Using (6) gives

$$\begin{aligned} C_{p^m}(a) &= \sum_{\substack{v \leq H^{1/8} \\ p^k | v}} \lambda_v + \sum_{j=0}^{k-1} \frac{G_{p^{m-2j}}(a)}{p^{m-2j}} \sum_{\substack{v \leq H^{1/8} \\ p^j \parallel v}} \lambda_v \\ &= b_{p^k} + \sum_{j=0}^{k-1} (b_{p^j} - b_{p^{j+1}}) \frac{G_{p^{m-2j}}(a)}{p^{m-2j}} + O(H^{-c/Y}). \end{aligned}$$

First suppose $p \leq Y^2$. Then $b_p = p^{-c_0/Y}$ is close to one, since

$$0 \leq 1 - b_p \leq \frac{c_0 \log p}{Y} \leq \frac{2c_0 \log Y}{Y}.$$

Since $b_{p^j} = b_p^j$, we have $b_{p^j} - b_{p^{j+1}} = b_{p^j}(1 - b_p)$. Using $m - 2j \geq k - j$ and setting $h = k - j$, the Gauss-sum bound gives

$$\begin{aligned} \frac{1}{b_{p^k}} \left| \sum_{j=0}^{k-1} (b_{p^j} - b_{p^{j+1}}) \frac{G_{p^{m-2j}}(a)}{p^{m-2j}} \right| &\leq \sqrt{2}(1 - b_p) \sum_{h=1}^k (\sqrt{p} b_p)^{-h} \\ &\leq \frac{\sqrt{2}(1 - b_p)}{\sqrt{p} b_p - 1} \ll \frac{c_0 \log p}{Y \sqrt{p}}. \end{aligned}$$

Here $1 - b_p \leq c_0 \log p / Y$ and $\sqrt{p} b_p - 1 \gg \sqrt{p}$ uniformly for $p \leq Y^2$ when Y is sufficiently large. Consequently,

$$C_{p^m}(a) = b_{p^k} \left(1 + O\left(\frac{c_0 \log p}{Y \sqrt{p}} \right) \right) + O(H^{-c/Y}),$$

uniformly in m , giving the desired estimate in this case.

Now suppose $p > Y^2$. For $0 \leq j < k$, the definition of b_d gives, for $0 \leq u \leq 1$,

$$b_{p^j} = c_0 Y \int_0^1 p^{-ku/(2 \log Y)} p^{(k-j)u/(2 \log Y)} u^{c_0 Y - 1} du \leq p^{(k-j)/(2 \log Y)} b_{p^k},$$

Using $0 \leq b_{p^j} - b_{p^{j+1}} \leq b_{p^j}$, $m - 2j \geq k - j = h$, the Gauss-sum bound gives

$$\begin{aligned} \frac{1}{b_{p^k}} \left| \sum_{j=0}^{k-1} (b_{p^j} - b_{p^{j+1}}) \frac{G_{p^{m-2j}}(a)}{p^{m-2j}} \right| &\leq \sqrt{2} \sum_{h=1}^k \left(p^{-1/2+1/(2 \log Y)} \right)^h \\ &\leq \frac{\sqrt{2} e / Y}{1 - e / Y} \ll \frac{1}{Y}. \end{aligned}$$

Here, for sufficiently large Y , $p^{-1/2+1/(2 \log Y)} \leq (Y^2)^{-1/2+1/(2 \log Y)} = e/Y$. As a consequence,

$$C_{p^m}(a) = b_{p^k} (1 + O(1/Y)) + O(H^{-c/Y}),$$

uniformly in m , thus giving the desired estimate in this case as well.

For general q , the proof is more subtle because b_d is not multiplicative. We instead use its representation as a weighted average of the multiplicative functions $\beta_d(u)$.

6 The complete sum

Recall that

$$C_q(a) = \frac{1}{q} \sum_{v \leq H^{1/8}} \lambda_v G_q(av^2). \quad (11)$$

The goal of this section is to show that, up to an exponentially small error, $C_q(a)$ lies in a fixed sector contained in the right half-plane for every $q \leq H$. As in Section 5, put

$$D = D(q) = \prod_{p^m \parallel q} p^{\lceil m/2 \rceil},$$

which is the least positive integer whose square is divisible by q . For each v divisible by D , we have $q \mid v^2$, so the phase av^2x^2/q is zero modulo 1. Hence the inner sum over x is trivially q , and so $G_q(av^2)/q = 1$. Hence the contribution to $C_q(a)$ from $v \leq H^{1/8}$ divisible by D is

$$\sum_{\substack{v \leq H^{1/8} \\ D \mid v}} \lambda_v = \begin{cases} b_D, & D \leq H^{1/8}, \\ 0, & D > H^{1/8}, \end{cases}$$

by (6). The following proposition shows that b_D is essentially dominant up to an exponentially small error.

Proposition 6.1. *Fix a sufficiently small absolute constant $c_0 > 0$, and let the parameters and weights be as in Section 5. Put $c = c_0/64$. Then for all sufficiently large H depending on c_0 , and uniformly for $1 \leq q \leq H$ and $a \in \mathbb{Z}$ with $(a, q) = 1$, we have*

$$C_q(a) = b_D(1 + O(c_0)) + O(H^{-c/Y}), \quad (12)$$

where the implied constants are absolute.

We first record some standard properties of the quadratic Gauss sums adapted to the present formulation.

Lemma 6.2. *Let $q \in \mathbb{N}$ and $a \in \mathbb{Z}$ with $(a, q) = 1$. Let $D = D(q)$ be as above. Then*

$$G_q(av^2) = G_q(a(v, D)^2) \quad (v \geq 1), \quad G_q(aD^2) = q.$$

In addition, we have

$$\frac{|G_q(ar^2)|}{q} \leq \sqrt{\frac{2r}{D}} \quad (13)$$

for every $r \mid D$.

Proof. The identity $G_q(aD^2) = q$ is trivial as $q \mid D^2$. For each $v \geq 1$, put

$$q_v = \frac{q}{(v^2, q)}, \quad a_v = \frac{av^2}{(v^2, q)}. \quad (14)$$

Then $(a_v, q_v) = 1$ and

$$\frac{G_q(av^2)}{q} = \frac{G_{q_v}(a_v)}{q_v}. \quad (15)$$

This reduction will also be used in Section 7.

Put $g = (v, D)$. Since $q \mid D^2$, we have

$$(g^2, q) = ((v, D)^2, q) = (v^2, (D^2, q)) = (v^2, q),$$

so that $q_g = q_v$. Moreover, setting $h = (v/g, q_v)$, we have $h^2g^2 \mid v^2$ and $h(v^2, q) \mid q$. Thus $h(g^2, (v^2, q)) \mid (v^2, q)$. Since $(v^2, q) \mid g^2$, we have $h(v^2, q) \mid (v^2, q)$, which implies $h = 1$. So $(v/g, q_v) = 1$. Consequently, since $a_v = a_g(v/g)^2$, changing variables by $x \mapsto (v/g)x$ in (15) gives

$$G_q(av^2) = q \cdot \frac{G_{q_v}(a_v)}{q_v} = q \cdot \frac{G_{q_v}(a_g)}{q_v} = q \cdot \frac{G_{q_g}(a_g)}{q_g} = G_q(ag^2).$$

Finally, by the square-root cancellation bound for quadratic Gauss sums:

$$\left| \frac{1}{q} \sum_{x \bmod q} e((ax^2 + bx)/q) \right| \leq \sqrt{\frac{2}{q}}, \quad (16)$$

where $q \in \mathbb{N}$, $a \in \mathbb{Z}$ with $(a, q) = 1$, and $b \in \mathbb{Z}$, we have, with $v = r \mid D$,

$$\frac{|G_q(ar^2)|}{q} = \frac{|G_{q_r}(a_r)|}{q_r} \leq \sqrt{\frac{2}{q_r}} \leq \sqrt{\frac{2r}{D}},$$

since $q_r \geq D(q_r) = D/g = D/r$. $\square$

The next lemma provides a useful approximation to $C_q(a)$ by a complete sum.

Lemma 6.3. *For all sufficiently large H depending on c_0 , and uniformly for $1 \leq q \leq H$ and $a \in \mathbb{Z}$ with $(a, q) = 1$, we have*

$$C_q(a) = \sum_{r \mid D} \frac{G_q(ar^2)}{q} \sum_{e \mid D/r} \mu(e) b_{re} + O(H^{-c/Y}), \quad (17)$$

where $c = c_0/64$.

Proof. For $q = 1$, both $C_1(a)$ and the main term in (17) equal $b_1 = 1$, so we may assume $q \geq 2$. For each $d \in \mathbb{N}$, write $d = d_1 d_2$ with d_1 and d_2 being the Y^2 -smooth and Y^2 -rough parts of d , respectively. For sufficiently large H , if $u \geq Y^{-1/2}$, then $u/(2 \log Y) \geq c_0/Y$ and therefore

$$\beta_d(u) = d_1^{-c_0/Y} d_2^{-u/(2 \log Y)} \leq d^{-c_0/Y}.$$

By definition (4),

$$b_d \leq c_0 Y \int_0^{Y^{-1/2}} u^{c_0 Y - 1} du + d^{-c_0/Y} c_0 Y \int_{Y^{-1/2}}^1 u^{c_0 Y - 1} du \leq Y^{-c_0 Y/2} + d^{-c_0/Y}. \quad (18)$$

Since $Y^2 = (\log H)/(\log \log H)$ and $\log Y \sim \frac{1}{2} \log \log H$, we have

$$\frac{Y}{2} \log Y \geq \frac{\log H}{8Y}$$

for sufficiently large H . Thus $b_d \leq 2H^{-c_0/(8Y)}$ whenever $d > H^{1/8}$.

We split the sum over $v \leq H^{1/8}$ in (11) according to the values of (v, D) . For each $r \mid D$, Möbius inversion gives

$$\mathbf{1}_{(v, D)=r} = \sum_{e \mid D/r} \mu(e) \mathbf{1}_{re \mid v}.$$

Plugging this in (11) and invoking Lemma 6.2 and (6) yields

$$C_q(a) = \sum_{r \mid D} \frac{G_q(ar^2)}{q} \sum_{v \leq H^{1/8}} \lambda_v \mathbf{1}_{(v, D)=r} = \sum_{r \mid D} \frac{G_q(ar^2)}{q} \sum_{\substack{e \mid D/r \\ re \leq H^{1/8}}} \mu(e) b_{re}. \quad (19)$$

We now remove the restriction $re \leq H^{1/8}$. For $re > H^{1/8}$, we have $b_{re} \leq 2H^{-c_0/(8Y)}$. Using (13) and then writing $h = D/r$, we see that the tail of (19) over $re > H^{1/8}$ is at most

$$\begin{aligned}
2H^{-c_0/(8Y)} \sum_{r|D} \frac{|G_q(ar^2)|}{q} \sum_{e|D/r} |\mu(e)| &\leq 2\sqrt{2}H^{-c_0/(8Y)} \sum_{h|D} \frac{1}{\sqrt{h}} \sum_{e|h} |\mu(e)| \\
&= 2\sqrt{2}H^{-c_0/(8Y)} \prod_{p^k \parallel D} \left(1 + 2 \sum_{j=1}^k p^{-j/2}\right) \\
&\leq 2\sqrt{2}H^{-c_0/(8Y)} \prod_{p|q} \left(1 + \frac{2}{\sqrt{p}-1}\right) \\
&\ll H^{-c_0/(8Y)} \exp\left(2 \sum_{p \leq p_{\omega(q)}} \frac{1}{\sqrt{p}-1}\right) \\
&= H^{-c_0/(8Y)} \exp\left(O\left(\sqrt{\omega(q) \log(3\omega(q))}\right)\right) \\
&= H^{-c_0/(8Y)} \exp\left(O\left(\sqrt{\log(3q)}\right)\right) \\
&\leq H^{-c_0/(8Y)} \exp(O(\sqrt{\log H})) \leq H^{-c/Y},
\end{aligned} \tag{20}$$

where $\omega(q)$ denotes the number of distinct prime factors of q and satisfies $\omega(q) \ll \log(3q)/\log \log(3q)$, and $p_{\omega(q)}$ is the $\omega(q)$ -th prime and satisfies $p_{\omega(q)} \ll \omega(q) \log(3\omega(q))$. Since

$$\frac{c_0 \log H}{8Y} = \frac{c_0}{8} \sqrt{\log H \log \log H},$$

the factor $\exp(O(\sqrt{\log H}))$ is absorbed with $c = c_0/64$ for sufficiently large H . This proves (17). $\square$

The proof of Proposition 6.1 now boils down to showing that the main contribution to $C_q(a)$ comes from the $r = D$ term in the complete sum in (17).

Proof of Proposition 6.1. For each $r \mid D$, complete multiplicativity of $\beta_d(u)$ gives

$$\sum_{e|D/r} \mu(e) \beta_{re}(u) = \beta_r(u) \sum_{e|D/r} \mu(e) \beta_e(u) = \beta_r(u) \prod_{p|D/r} (1 - \beta_p(u)),$$

whence

$$\sum_{e|D/r} \mu(e) b_{re} = c_0 Y \int_0^1 \beta_r(u) \prod_{p|D/r} (1 - \beta_p(u)) u^{c_0 Y - 1} du \geq 0. \tag{21}$$

Thus the completed sum in Lemma 6.3 has nonnegative coefficients, even though the original λ_v may have either sign. The term with $r = D$ is exactly b_D , since $G_q(aD^2) = q$. Applying (13) to the remaining terms and writing $h = D/r > 1$, we obtain

$$|C_q(a) - b_D| \leq \sqrt{2} c_0 Y \int_0^1 \beta_D(u) \sum_{\substack{h|D \\ h>1}} \frac{1}{\sqrt{h} \beta_h(u)} \prod_{p|h} (1 - \beta_p(u)) u^{c_0 Y - 1} du + O(H^{-c/Y}). \tag{22}$$

For sufficiently large H , both c_0/Y and $1/(2 \log Y)$ are at most $1/4$, so $\beta_p(u) \geq p^{-1/4}$ and

$\sqrt{p}\beta_p(u) \geq p^{1/4} > 1$. In conjunction with complete multiplicativity of $\beta_d(u)$, these inequalities imply

$$\begin{aligned} \sum_{\substack{h|D \\ h>1}} \frac{1}{\sqrt{h}\beta_h(u)} \prod_{p|h} (1 - \beta_p(u)) &\leq \prod_{p|D} \left(1 + (1 - \beta_p(u)) \sum_{j=1}^{\infty} (\sqrt{p}\beta_p(u))^{-j} \right) - 1 \\ &= \prod_{p|D} \left(1 + \frac{1 - \beta_p(u)}{\sqrt{p}\beta_p(u) - 1} \right) - 1 \\ &\leq \exp \left(\sum_{p|D} \frac{1 - \beta_p(u)}{\sqrt{p}\beta_p(u) - 1} \right) - 1. \end{aligned} \quad (23)$$

We split the prime factors $p \mid D$ according to their sizes relative to Y^2 . For $0 \leq s \leq 1/4$, the inequalities

$$1 - p^{-s} \leq s \log p, \quad p^{1/2-s} - 1 \geq (1 - 2^{-1/4})p^{1/2-s}$$

give

$$\frac{1 - p^{-s}}{p^{1/2-s} - 1} \ll s(\log p)p^{s-1/2},$$

with an absolute implicit constant. If $p \leq Y^2$, then $\beta_p(u) = p^{-s}$ with $s = c_0/Y$. Since $Y^{2c_0/Y} = \exp(2c_0 \log Y/Y) = O(1)$, we get

$$\frac{1 - \beta_p(u)}{\sqrt{p}\beta_p(u) - 1} \ll \frac{c_0 \log p}{Y\sqrt{p}}.$$

Summing this over $p \leq Y^2$ using Chebyshev's estimate and partial summation, we obtain

$$\sum_{\substack{p|D \\ p \leq Y^2}} \frac{1 - \beta_p(u)}{\sqrt{p}\beta_p(u) - 1} \ll \frac{c_0}{Y} \sum_{p \leq Y^2} \frac{\log p}{\sqrt{p}} \ll c_0.$$

Now consider the remaining primes, namely those with $p > Y^2$. We apply the same inequalities but with $s = u/(2 \log Y)$. The exponent $u/(2 \log Y) - 1/2$ is negative, so

$$p^{u/(2 \log Y) - 1/2} \leq (Y^2)^{u/(2 \log Y) - 1/2} = \frac{e^u}{Y} \leq \frac{e}{Y}.$$

As a result,

$$\frac{1 - \beta_p(u)}{\sqrt{p}\beta_p(u) - 1} \ll \frac{u \log p}{Y \log Y}.$$

Write $D = D_1 D_2$ with D_1 and D_2 being the Y^2 -smooth and Y^2 -rough parts of D , respectively. Then

$$\sum_{\substack{p|D \\ p > Y^2}} \frac{1 - \beta_p(u)}{\sqrt{p}\beta_p(u) - 1} \ll \frac{u}{Y \log Y} \sum_{\substack{p|D \\ p > Y^2}} \log p \leq \frac{u \log D_2}{Y \log Y}.$$

Gathering the estimates above and returning to (23), we have

$$\sum_{\substack{h|D \\ h>1}} \frac{1}{\sqrt{h}\beta_h(u)} \prod_{p|h} (1 - \beta_p(u)) \leq \exp \left(C \left(c_0 + \frac{u \log D_2}{2Y \log Y} \right) \right) - 1$$

for some absolute constant $C > 0$. Since

$$\beta_D(u) = D_1^{-c_0/Y} D_2^{-u/(2 \log Y)},$$

it follows that

$$\beta_D(u) \sum_{\substack{h|D \\ h>1}} \frac{1}{\sqrt{h} \beta_h(u)} \prod_{p|h} (1 - \beta_p(u)) \leq e^{C c_0} \beta_D((1 - C/Y)u) - \beta_D(u).$$

Inserting this into (22) gives

$$|C_q(a) - b_D| \leq \sqrt{2} e^{C c_0} c_0 Y \int_0^1 \beta_D((1 - C/Y)u) u^{c_0 Y - 1} du - \sqrt{2} b_D + O(H^{-c/Y}).$$

For sufficiently large H , we have $0 < C/Y < 1/2$. Changing variables by $w = (1 - C/Y)u$, we have

$$\begin{aligned} c_0 Y \int_0^1 \beta_D((1 - C/Y)u) u^{c_0 Y - 1} du &= \frac{c_0 Y}{(1 - C/Y)^{c_0 Y}} \int_0^{1-C/Y} \beta_D(w) w^{c_0 Y - 1} dw \\ &\leq (1 - C/Y)^{-c_0 Y} b_D. \end{aligned}$$

Therefore,

$$|C_q(a) - b_D| \leq \sqrt{2} b_D (e^{C c_0} (1 - C/Y)^{-c_0 Y} - 1) + O(H^{-c/Y}) \ll c_0 b_D + H^{-c/Y},$$

since

$$e^{C c_0} (1 - C/Y)^{-c_0 Y} - 1 = (1 + O(C c_0)) \left(1 + O\left(c_0 Y \cdot \frac{C}{Y}\right) \right) - 1 \ll c_0.$$

This proves (12). □

7 Applying the Poisson summation formula

We use a standard Poisson summation argument to prove the estimate (10) for $\mathcal{B}_H(\theta)$.

Proposition 7.1. *Let $1 \leq q \leq H$, $(a, q) = 1$, and*

$$\theta = \frac{a}{q} + \eta, \quad |\eta| \leq \frac{2}{qH}.$$

Then

$$\mathcal{B}_H(\theta) = C_q(a) \int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx + O(H^{3/4}), \quad (24)$$

where the implied constant is absolute.

Proof. Recall the definition of $\mathcal{B}_H$ from equation (7):

$$\mathcal{B}_H(\theta) = \sum_{v \leq H^{1/8}} v \lambda_v \sum_{\substack{x \in \mathbb{Z} \\ 0 < v^2 x^2 \leq N}} e^{-v^2 x^2/H^2} e(\theta v^2 x^2). \quad (25)$$

Since $0 < b_d \leq 1$, the definition (5) gives

$$|\lambda_v| \leq H^{1/8}/v, \quad \sum_{v \leq H^{1/8}} v |\lambda_v| \leq H^{1/4}. \quad (26)$$

We first restore $x = 0$ and remove the cutoff $v^2 x^2 \leq N$. Restoring $x = 0$ changes the sum by at most

$$\sum_{v \leq H^{1/8}} v \lambda_v = O(H^{1/4}).$$

The contribution from the tail given by $v^2x^2 > N$ is also negligible. Indeed, if $v^2x^2 > N$, then

$$e^{-v^2x^2/H^2} \leq e^{-N/(2H^2)} e^{-v^2x^2/(2H^2)}.$$

Since $v \leq H^{1/8}$, an integral comparison gives $\sum_{x \in \mathbb{Z}} e^{-v^2x^2/(2H^2)} \ll H/v$. Hence the tail contribution is at most

$$e^{-N/(2H^2)} \sum_{v \leq H^{1/8}} v |\lambda_v| \sum_{x \in \mathbb{Z}} e^{-v^2x^2/(2H^2)} \ll H^{5/4} e^{-(\log N)^2/2} \ll 1,$$

where we used $N/H^2 = (\log N)^2$ and (26). Thus

$$\mathcal{B}_H(\theta) = \sum_{v \leq H^{1/8}} v \lambda_v \sum_{x \in \mathbb{Z}} e^{-v^2x^2/H^2} e(\theta v^2x^2) + O(H^{1/4}).$$

Fix $v \leq H^{1/8}$ and recall from (14) that

$$q_v = \frac{q}{(v^2, q)}, \quad a_v = \frac{av^2}{(v^2, q)}, \quad \frac{av^2}{q} = \frac{a_v}{q_v}, \quad (a_v, q_v) = 1.$$

Define

$$\Phi_v(x) = e^{-v^2x^2/H^2} e(\eta v^2x^2).$$

This belongs to $\mathcal{S}(\mathbb{R})$, since each derivative is a polynomial times Φ_v and $|\Phi_v(x)| = e^{-v^2x^2/H^2}$. On the residue class $x \equiv r \pmod{q_v}$, the rational phase $e(a_vx^2/q_v)$ equals $e(a_vr^2/q_v)$. Splitting into these classes and applying (2), we obtain

$$\begin{aligned} \sum_{x \in \mathbb{Z}} e(a_vx^2/q_v) \Phi_v(x) &= \sum_{r \bmod q_v} e(a_vr^2/q_v) \sum_{\substack{x \in \mathbb{Z} \\ x \equiv r \pmod{q_v}}} \Phi_v(x) \\ &= \frac{1}{q_v} \sum_{\ell \in \mathbb{Z}} \left(\sum_{r \bmod q_v} e\left(\frac{a_vr^2 + \ell r}{q_v}\right) \right) \widehat{\Phi}_v\left(\frac{\ell}{q_v}\right). \end{aligned} \tag{27}$$

The zero frequency gives the main term. Indeed,

$$\frac{1}{q_v} \sum_{r \bmod q_v} e(a_vr^2/q_v) = \frac{G_{q_v}(a_v)}{q_v} = \frac{G_q(av^2)}{q}$$

by (15). Also, changing variables by $y = vx$ gives

$$v \widehat{\Phi}_v(0) = v \int_{\mathbb{R}} e^{-v^2x^2/H^2} e(\eta v^2x^2) dx = \int_{\mathbb{R}} e^{-y^2/H^2} e(\eta y^2) dy.$$

Thus, after multiplication by v , the contribution from $\ell = 0$ in (27) is

$$\frac{G_q(av^2)}{q} \int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx.$$

It remains to bound the contribution from $\ell \neq 0$. By the formula for the Fourier transform of the Gaussian, we have

$$|\widehat{\Phi}_v(\xi)| = \frac{\sqrt{\pi}H}{v(1 + 4\pi^2\eta^2H^4)^{1/4}} \exp\left(-\frac{\pi^2H^2\xi^2}{v^2(1 + 4\pi^2\eta^2H^4)}\right). \tag{28}$$

Since $(a_v, q_v) = 1$, the Gauss-sum bound (16) gives, for every $\ell \in \mathbb{Z}$,

$$\left| \frac{1}{q_v} \sum_{r \bmod q_v} e\left(\frac{a_vr^2 + \ell r}{q_v}\right) \right| \leq \sqrt{\frac{2}{q_v}}.$$

Therefore the total contribution from $\ell \neq 0$, after multiplication by v , is at most

$$\frac{\sqrt{2\pi} H}{\sqrt{q_v}(1 + 4\pi^2\eta^2 H^4)^{1/4}} \sum_{\ell \neq 0} \exp\left(-\frac{\pi^2 H^2 \ell^2}{v^2 q_v^2 (1 + 4\pi^2\eta^2 H^4)}\right).$$

For every $A > 0$, monotonicity of the Gaussian gives

$$\sum_{\ell \neq 0} e^{-A^2 \ell^2} = 2 \sum_{\ell \geq 1} e^{-A^2 \ell^2} \leq 2 \int_0^\infty e^{-A^2 x^2} dx = \frac{\sqrt{\pi}}{A}.$$

Applying this with $A = \pi H / (v q_v \sqrt{1 + 4\pi^2\eta^2 H^4})$ bounds the nonzero frequencies by

$$\ll v \sqrt{q_v} (1 + 4\pi^2\eta^2 H^4)^{1/4}.$$

Combining the zero and nonzero frequencies, we have proved

$$v \sum_{x \in \mathbb{Z}} e(a_v x^2 / q_v) \Phi_v(x) = \frac{G_q(av^2)}{q} \int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx + O\left(v \sqrt{q_v} (1 + 4\pi^2\eta^2 H^4)^{1/4}\right).$$

We now sum against λ_v . The integral is independent of v , and $q^{-1} \sum_v \lambda_v G_q(av^2) = C_q(a)$, so the main terms give the one in (24). For the errors, use $q_v \leq q$ and

$$q \sqrt{1 + 4\pi^2\eta^2 H^4} \leq q + 2\pi q |\eta| H^2 \leq (1 + 4\pi) H.$$

It follows from (26) that their sum is at most

$$\sqrt{q} (1 + 4\pi^2\eta^2 H^4)^{1/4} \sum_{v \leq H^{1/8}} v |\lambda_v| \ll H^{1/2} H^{1/4} = H^{3/4}.$$

This error absorbs the $O(H^{1/4})$ error from restoring $x = 0$ and removing the cutoff $x^2 v^2 \leq N$, proving (24). $\square$

8 Finishing the argument

We now combine Propositions 6.1 and 7.1 to prove Proposition 5.1, which is the uniform lower bound for $\mathcal{B}_H(\theta)$.

Proof of Proposition 5.1. For every $\eta \in \mathbb{R}$, the Gaussian integral evaluates to

$$\int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx = H \int_{\mathbb{R}} \exp(-(1 - 2\pi i \eta H^2) y^2) dy = \frac{\sqrt{\pi} H}{\sqrt{1 - 2\pi i \eta H^2}}, \quad (29)$$

where the square root is chosen to have positive real part.

Taking $a = 0$, $q = 1$, and $\eta = 0$ in Proposition 7.1, and using $C_1(0) = \sum_{v \leq H^{1/8}} \lambda_v = b_1 = 1$, gives

$$\mathcal{B}_H(0) = \sqrt{\pi} H + O(H^{3/4}).$$

This proves (8). In particular, $\mathcal{B}_H(0) > 0$ and $\mathcal{B}_H(0) \asymp H$ for sufficiently large H .

Fix $\theta \in \mathbb{R}$. Dirichlet approximation, with denominator bound $\lfloor H \rfloor$, supplies coprime integers a, q such that

$$1 \leq q \leq H, \quad \theta = \frac{a}{q} + \eta, \quad |\eta| \leq \frac{2}{qH}.$$

Propositions 6.1 and 7.1 give

$$\mathcal{B}_H(\theta) = b_D(1 + O(c_0)) \int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx + O(H^{1-c/Y} + H^{3/4}),$$

where $D = D(q)$. Here we used (29) to bound the modulus of the integral by $\sqrt{\pi}H$.

We now check that the two factors in the main term lie in sectors whose product has positive real part. By (29),

$$\arg\left(\int_{\mathbb{R}} e^{-x^2/H^2} e(\eta x^2) dx\right) = -\frac{1}{2} \arg(1 - 2\pi i \eta H^2) \in (-\pi/4, \pi/4).$$

Since $b_D > 0$ and the implied constant in $O(c_0)$ is absolute, choosing c_0 sufficiently small also gives

$$\arg(b_D(1 + O(c_0))) \in (-\pi/4, \pi/4).$$

Their product therefore has argument in $(-\pi/2, \pi/2)$ and hence positive real part. It follows that

$$\operatorname{Re} \mathcal{B}_H(\theta) \geq -C(H^{1-c/Y} + H^{3/4}).$$

Since $H^{3/4} \leq H^{1-c/Y}$ for sufficiently large H , increasing C if necessary gives

$$\operatorname{Re} \mathcal{B}_H(\theta) \geq -CH^{1-c/Y} \quad (\theta \in \mathbb{R}).$$

Together with (8), this proves the proposition. $\square$

A The improved bound for the Furstenberg–Sárközy theorem

We record the standard proof that the van der Corput property implies the Sárközy-type result promised in the introduction. For this we use Fourier analysis on the finite cyclic group $\mathbb{Z}/M\mathbb{Z}$, with the convention in Section 4.

Proof of Corollary 1.2. Let $M = 2N + 1$, and regard A as a subset of $\mathbb{Z}/M\mathbb{Z}$. We apply Theorem 1.1 to obtain a trigonometric polynomial

$$P(\theta) = a_0 + \sum_{1 \leq n \leq \sqrt{N}} a_n \cos(2\pi n^2 \theta)$$

such that $P(\theta) \geq 0$ for every $\theta \in \mathbb{R}/\mathbb{Z}$, and $P(0) = 1$. Let

$$\widehat{1_A}(\xi) = \sum_{a \in A} e(a\xi/M), \quad \xi \in \mathbb{Z}/M\mathbb{Z}.$$

Since $A \subseteq [N]$, $1 \leq d \leq N$, and $M = 2N + 1$, the congruence $a' - a \equiv d \pmod{M}$ is the same as the equality $a' - a = d$. Hence the assumption that $A - A$ contains no nonzero square gives

$$\frac{1}{M} \sum_{\xi \pmod{M}} |\widehat{1_A}(\xi)|^2 P(\xi/M) = \frac{a_0}{M} \sum_{\xi \pmod{M}} |\widehat{1_A}(\xi)|^2 = a_0 |A|.$$

Taking real parts and separating the zero frequency, we get

$$a_0 |A| = \frac{|A|^2}{M} + \frac{1}{M} \sum_{\substack{\xi \pmod{M} \\ \xi \neq 0}} |\widehat{1_A}(\xi)|^2 P(\xi/M) \geq \frac{|A|^2}{M},$$

since $P(0) = 1$ and every remaining summand is nonnegative. This implies

$$|A| \leq a_0 M \ll N \exp(-c\sqrt{\log N \log \log N}),$$

as desired. $\square$

B Proof of the lower bound

We close our discussion with a proof of Theorem 1.3. The critical input is the following form of Harnack's inequality for harmonic functions; see [BS06, Lemma 2.5].

Lemma B.1 (Harnack's inequality). *If h is a nonnegative harmonic function defined on the right half-plane, then for all real numbers $0 < a \leq b$,*

$$h(b) \geq \frac{a}{b} h(a).$$

We need the following version of Harnack's inequality for harmonic functions on a sector contained in the right half-plane.

Corollary B.2. *Let $\kappa \geq 1$. If h is a nonnegative harmonic function on the sector*

$$\{w \in \mathbb{C} \setminus \{0\} : |\arg w| < \pi/(2\kappa)\},$$

then, for all real numbers $0 < a \leq b$,

$$h(b) \geq \left(\frac{a}{b}\right)^\kappa h(a). \quad (30)$$

Proof. This follows immediately from Lemma B.1 applied to the nonnegative harmonic function $z \mapsto h(z^{1/\kappa})$ on the right half-plane at the positive real points $a^\kappa \leq b^\kappa$. $\square$

For a prime p , let $\nu_p(n)$ denote the p -adic valuation of a positive integer n . In other words, $\nu_p(n)$ is the largest nonnegative integer k such that $p^k \mid n$.

Lemma B.3. *Let $p \equiv 3 \pmod{4}$ be prime. If $\operatorname{Re} w > 0$ and $|\operatorname{Im} w| \leq (\operatorname{Re} w)/(2\sqrt{p})$, there is a probability measure $\mu_{p,w}$ on $\mathbb{T}$ such that*

$$\int_{\mathbb{T}} e(n^2 \alpha) d\mu_{p,w}(\alpha) = \exp\left(-2^{-\nu_p(n)} w\right) \quad (n \geq 1), \quad (31)$$

and

$$\mu_{p,w}(\{0\}) \geq e^{-\operatorname{Re} w}.$$

We first deduce Theorem 1.3 from this lemma, and then prove the lemma.

Proof of Theorem 1.3 assuming Lemma B.3. Put $X = \lfloor \sqrt{N} \rfloor$ and fix a polynomial

$$P(\alpha) = a_0 + \sum_{1 \leq n \leq X} a_n \cos(2\pi n^2 \alpha), \quad P \geq 0, \quad P(0) = 1.$$

Orthogonality and nonnegativity give

$$a_0 = \int_{\mathbb{T}} P(\alpha) d\alpha > 0, \quad |a_n| = 2 \left| \int_{\mathbb{T}} P(\alpha) e(-n^2 \alpha) d\alpha \right| \leq 2a_0. \quad (32)$$

We may assume that X is sufficiently large and $a_0 \leq 1$. Let

$$y = \frac{4 \log X}{\log \log X},$$

and denote by $\mathcal{P}$ the set of primes $p \leq y$ such that $p \equiv 3 \pmod{4}$. The prime number theorem for primes in the fixed progression $3 \pmod{4}$ gives

$$\sum_{p \in \mathcal{P}} \log p = (1 + o(1)) \frac{y}{2} \geq \frac{y}{4} = \frac{\log X}{\log \log X}.$$

For every $1 \leq n \leq X$,

$$\left(\min_{p \in \mathcal{P}} \nu_p(n) \right) \sum_{p \in \mathcal{P}} \log p \leq \sum_{p \in \mathcal{P}} \nu_p(n) \log p \leq \log n \leq \log X.$$

Thus some $p \in \mathcal{P}$ has $\nu_p(n) \leq \log \log X$, and hence

$$M(n) := \sum_{p \in \mathcal{P}} 2^{-\nu_p(n)} \geq 2^{-\log \log X} \geq \frac{1}{\log X}. \quad (33)$$

For every $p \in \mathcal{P}$ and every $w \in \mathbb{C}$ with $\operatorname{Re} w > 0$ and $|\operatorname{Im} w| \leq (\operatorname{Re} w)/(2\sqrt{y})$, Lemma B.3 supplies a probability measure $\mu_{p,w}$ on $\mathbb{T}$. Set

$$\mu_w := \ast_{p \in \mathcal{P}} \mu_{p,w},$$

the convolution of all $\mu_{p,w}$ over $p \in \mathcal{P}$. Then

$$\int_{\mathbb{T}} e(n^2 \alpha) d\mu_w(\alpha) = \prod_{p \in \mathcal{P}} \exp(-2^{-\nu_p(n)} w) = e^{-M(n)w}. \quad (34)$$

Consequently, the entire function

$$F(w) := a_0 + \sum_{1 \leq n \leq X} a_n e^{-M(n)w}$$

satisfies

$$\operatorname{Re} F(w) = \int_{\mathbb{T}} P(\alpha) d\mu_w(\alpha) \geq 0 \quad (35)$$

for every $w \in \mathbb{C}$ with $\operatorname{Re} w > 0$ and $|\operatorname{Im} w| \leq (\operatorname{Re} w)/(2\sqrt{y})$. For any real $s > 0$,

$$F(s) \geq \mu_s(\{0\})P(0) \geq \prod_{p \in \mathcal{P}} \mu_{p,s}(\{0\}) \geq \prod_{p \in \mathcal{P}} e^{-s} = e^{-s|\mathcal{P}|}. \quad (36)$$

Put $b = 2(\log X)^2$. By (32) and (33),

$$|F(b) - a_0| \leq 2a_0 X e^{-b/\log X} = \frac{2a_0}{X} \leq a_0,$$

so $F(b) \leq 2a_0$. Set

$$\kappa = \frac{\pi}{2 \arctan(1/(2\sqrt{y}))} \asymp \sqrt{y}.$$

The interior of the sector in (35) is $|\arg w| < \pi/(2\kappa)$. Applying Corollary B.2 to $\operatorname{Re} F$ at $1/|\mathcal{P}|$ and b , and then using (36), we have

$$2a_0 \geq F(b) \geq (b|\mathcal{P}|)^{-\kappa} F(1/|\mathcal{P}|) \geq e^{-1} (b|\mathcal{P}|)^{-\kappa}. \quad (37)$$

Since $|\mathcal{P}| \leq y$ and $\log(by) \ll \log \log X$, it follows that

$$\log \frac{1}{a_0} \ll \sqrt{y} \log \log X \ll \sqrt{\log X \log \log X} \ll \sqrt{\log N \log \log N},$$

which proves the theorem. $\square$

We now show the existence of the required probability measures $\mu_{p,w}$.

Proof of Lemma B.3. Write $s = \operatorname{Re} w > 0$ and put

$$\tau = -\frac{2i \operatorname{Im} w}{s}, \quad |\tau| \leq \frac{1}{\sqrt{p}}.$$

For each $a \in \mathbb{Z}/p\mathbb{Z}$, define

$$q_a = \frac{1}{p} + \frac{\operatorname{Im} \tau}{\sqrt{p}} \left(\frac{a}{p} \right),$$

where $\left(\frac{a}{p} \right)$ denotes the Legendre symbol. These coefficients are nonnegative and sum to one. For each $j \geq 0$, let

$$\sigma_{j,\tau} = \frac{1}{p^{2j}} \sum_{b \in \mathbb{Z}/p^{2j}\mathbb{Z}} \sum_{a \in \mathbb{Z}/p\mathbb{Z}} q_a \delta_{(a+pb)/p^{2j+1}},$$

where δ_x denotes the Dirac measure at $x \in \mathbb{T}$. This is a finitely supported probability measure. Orthogonality gives

$$\frac{1}{p^{2j}} \sum_{b \in \mathbb{Z}/p^{2j}\mathbb{Z}} e\left(\frac{n^2 b}{p^{2j}}\right) = \mathbf{1}_{p^j | n},$$

so

$$\int_{\mathbb{T}} e(n^2 \alpha) d\sigma_{j,\tau}(\alpha) = 0$$

when $p^j \nmid n$. If $n = p^j r$ with $p \nmid r$, the quadratic Gauss sum

$$\sum_{a \in \mathbb{Z}/p\mathbb{Z}} \left(\frac{a}{p} \right) e\left(\frac{ar^2}{p}\right) = \sum_{x \in \mathbb{Z}/p\mathbb{Z}} e\left(\frac{x^2}{p}\right) = i\sqrt{p}$$

implies

$$\int_{\mathbb{T}} e(n^2 \alpha) d\sigma_{j,\tau}(\alpha) = \sum_{a \in \mathbb{Z}/p\mathbb{Z}} q_a e\left(\frac{ar^2}{p}\right) = i \operatorname{Im} \tau = \tau.$$

If $p \mid r$, every exponential in this sum is one, so that

$$\int_{\mathbb{T}} e(n^2 \alpha) d\sigma_{j,\tau}(\alpha) = \begin{cases} 0, & \nu_p(n) < j, \\ \tau, & \nu_p(n) = j, \\ 1, & \nu_p(n) > j. \end{cases} \quad (38)$$

Now consider the probability measure

$$\rho = \sum_{j=0}^{\infty} 2^{-j-1} \sigma_{j,\tau}.$$

For $k = \nu_p(n)$, we obtain

$$\int_{\mathbb{T}} e(n^2 \alpha) d\rho(\alpha) = \sum_{j=0}^{k-1} 2^{-j-1} + 2^{-k-1} \tau = 1 - 2^{-k} \frac{w}{s}.$$

Define

$$\mu_{p,w} = e^{-s} \sum_{r=0}^{\infty} \frac{s^r}{r!} \rho^{*r},$$

where

$$\rho^{*r} := \underbrace{\rho * \cdots * \rho}_{r \text{ times}}$$

with the convention that $\rho^{*0} = \delta_0$. Then $\mu_{p,w}$ is a probability measure. Moreover,

$$\int_{\mathbb{T}} e(n^2\alpha) d\mu_{p,w}(\alpha) = e^{-s} \sum_{r=0}^{\infty} \frac{s^r}{r!} \left(1 - 2^{-k} \frac{w}{s}\right)^r = \exp\left(-s + s \left(1 - 2^{-k} \frac{w}{s}\right)\right) = e^{-2^{-k}w}.$$

Finally, the $r = 0$ term gives $\mu_{p,w}(\{0\}) \geq e^{-s}$. □

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